Mars Habitat Automation Platform

Laboratory of Advanced Programming

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Project Context

Sensors

- Temperature
- Humidity
- Pressure
- Air Quality

→

Monitoring Platform

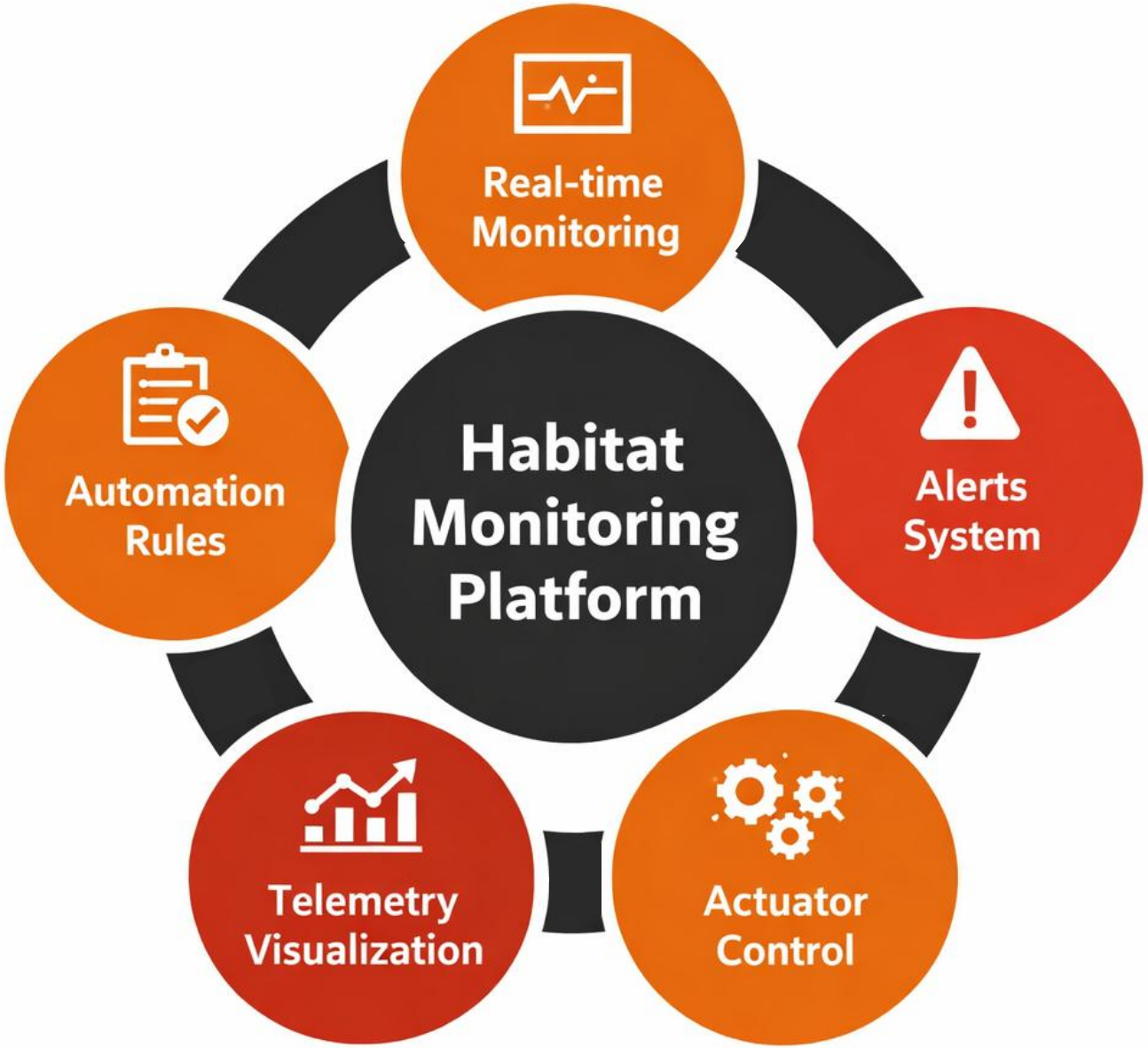
Real-time habitat control

→

Actuators

- Cooling Fans
- Heaters
- Ventilation

Platform Capabilities



Technology Stack

Frontend

- React
- Vite
- WebSocket
- Recharts

Backend

- FastAPI (Python)
- asyncio
- aiokafka
- SQLAlchemy
- PostgreSQL

Data

- PostgreSQL database

Infrastructure

- Docker Compose
- Kafka
- PostgreSQL

Communication

- Kafka topics
- HTTP APIs
- WebSocket connections

Microservices Overview

Ingestion Service

Collects sensor data from the IoT simulator

Processing Service

Normalizes and validates sensor events

Rule Management Service

Stores and manages automation rules

Actuator Rules Service

Evaluates rules and triggers actions

Microservices Overview

Realtime Service

Pushes real-time updates to the dashboard

Actuator Management Service

Controls actuators through simulator APIs

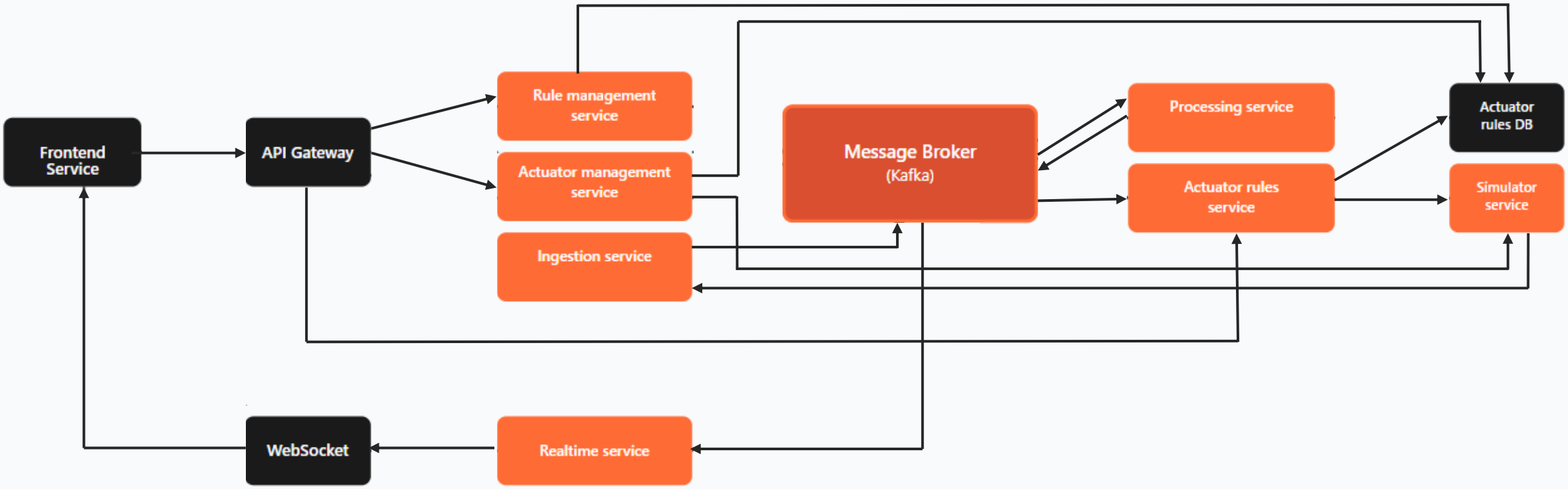
API Gateway

Single entry point for frontend requests

Frontend Service

Web dashboard user interface

System Architecture



System Demonstration

Dashboard Monitoring

- Real-time sensor data
- Live telemetry charts
- System health status
- Historical trends

Automation Rule Creation

- Define conditions
- Set thresholds
- Configure actions
- Rule validation

Actuator Activation

- Manual control
- Automated triggers
- Status feedback
- Activity logging